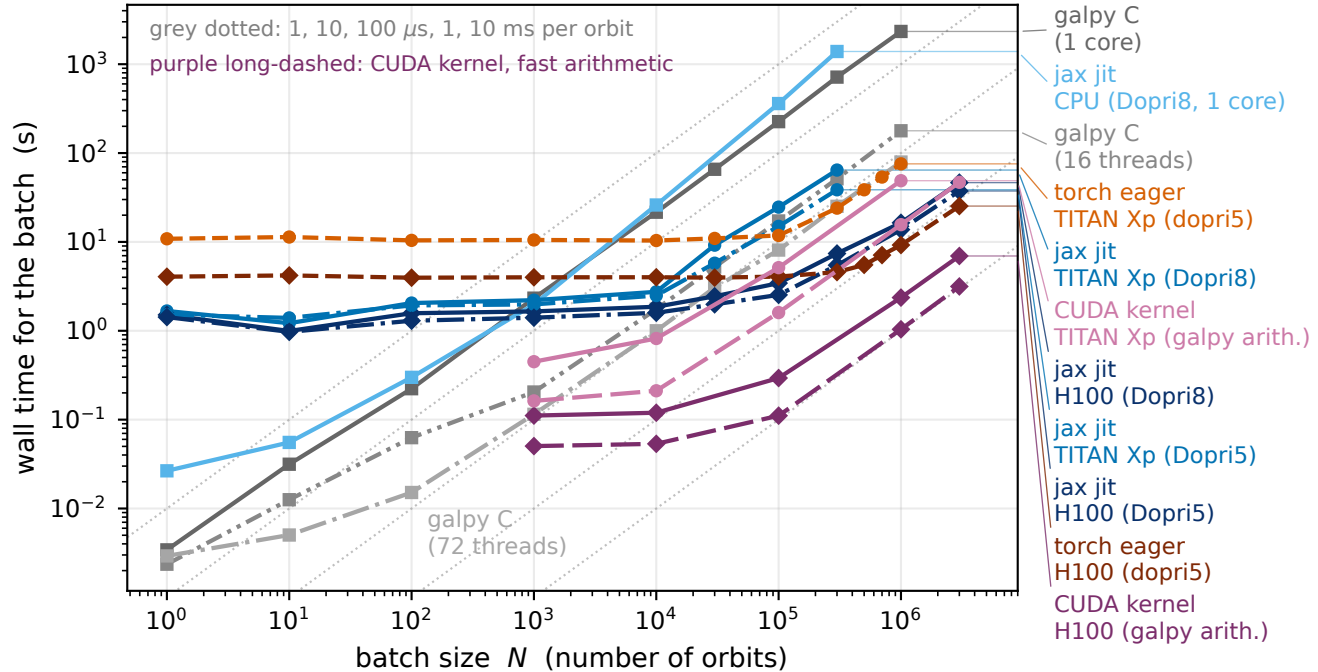
grey dotted: 1, 10, 100 μs, 1, 10 ms per orbit
purple long-dashed: CUDA kernel, fast arithmetic
wall time for the batch (s)
batch size N (number of orbits)
galpy C
(72 threads)
galpy C
(1 core)
jax jit
CPU (Dopri8, 1 core)
galpy C
(16 threads)
torch eager
TITAN Xp (dopri5)
jax jit
TITAN Xp (Dopri8)
CUDA kernel
TITAN Xp (galpy arith.)
jax jit
H100 (Dopri8)
jax jit
TITAN Xp (Dopri5)
jax jit
H100 (Dopri5)
torch eager
H100 (dopri5)
CUDA kernel
H100 (galpy arith.)